

James Tribble

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EDUCATION

Clemson University

Bachelor of Science in Computer Science, Minors in Math and Artificial Intelligence

GPA: 4.0/4.0

Clemson, SC

May 2027

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript

Tools & Frameworks: AWS, Git, GitHub, GitHub Copilot, Docker, Linux, Microsoft Office

Libraries: PyTorch, OpenCV, Pandas, NumPy, Scikit-Learn, Matplotlib

Relevant Skills: Linear Algebra, Differential Equations, Statistics, Probability Theory, Computer Vision, Artificial Intelligence, Machine Learning, Data Science, Agile, DevOps

Soft Skills: Teaching, Time management, Technical Assistance

EXPERIENCE

Honors Thesis - Undergraduate Research Assistant

Clemson University, Vision and Learning Lab

August 2025 - present

Clemson, SC

- 2nd author: "Motion Focus Recognition in Fast-Moving Egocentric Video" (WACV)
- Created a 637 Q&A pairs for an egocentric video benchmark

MemoryScapes - Computer Vision Engineer

Clemson University, Senior Capstone

August 2025 - Dec 2025

Clemson, SC

- Deployed experimental Nightshade research code using Docker on AWS EC2
- Scripted automated poisoning of public MemoryScapes user photos
- Evaluated nightshade by poisoning 6 SDXL LoRAs trained on user data

Rural Robotics REU - Undergraduate Research Assistant

West Virginia University, Interactive Robotics Laboratory

May 2025 - July 2025

Morgantown, WV

- Developed an iOS App that leveraged ARKit and LiDAR to collect known Camera Poses and images.
- Wrote Bash scripts to automate data preparation for 4 Novel View Synthesis models
- Evaluated 4 Novel View Synthesis algorithms on 3 outdoor scenes, and found that SVRaster performed the best
- Set up Docker containers to simulate autonomous robot navigation in Gazebo

Teachers Assistant

Clemson University, School of Computing

August 2024 - May 2025

Clemson, SC

- Served as a Teaching Assistant for Data Structures (31 students) and Computer Organization (32 students), leading weekly lab sessions and office hours for each course

NIMBUS Lab REU - Undergraduate Research Assistant

University of Nebraska Lincoln

June 2024 – August 2024

Lincoln, NE

- Utilized 1.1 gigabytes of data from an intersection traffic video to train multiple reinforcement learning models to navigate a UAV to a position that mitigated propagation delay between an attached base station and user equipment

PROJECTS

Policy Document Validation | RAG, NLP

Spring 2026

- Won **Most Viable Product** at LPL Financial's 2026 Hackathon
- Extracted and structured data from multi-document uploads using AWS Textract
- Validated compliance policies using AWS Bedrock Knowledge Bases
- Developed React interface for real-time document processing updates

South Carolina Legislation Retrieval | RAG, NLP

Spring 2025

- Grouped 1,500+ South Carolina legislative bills into topics using k-means clustering (elbow method for optimal k), then trained a gradient boosting classifier on these labels, achieving 90% F1-score in predicting bill categories
- Built a legislative chatbot using Retrieval-Augmented Generation to query and summarize 2,500+ laws, enabling natural language access to legal documents